

System Architecture

XBandDigitalModule_20x20 - System Architecture Document

Document Information

Field	Value
Document ID	XBDM-DOC-001
Version	1.0
Date	2026-09-09
Classification	ITAR Restricted
Status	Preliminary Des

1. Executive Summary

The XBandDigitalModule_20x20 is a 20cm x 20cm space-grade electronic module designed for RF signal acquisition and sampling up to 500 MHz bandwidth in X-band (8-12 GHz). The module implements a direct

2. System Overview

2.1 Mission Requirements

- **Frequency Range**: 8-12 GHz (X-band)

2.2 Design Philosophy

The module?? direct RF sampling architecture to:

3. Architecture Selection

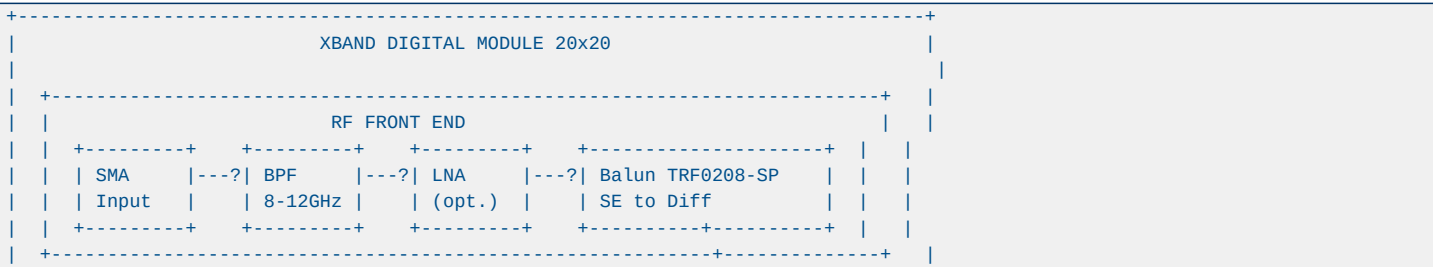
3.1 Options Considered

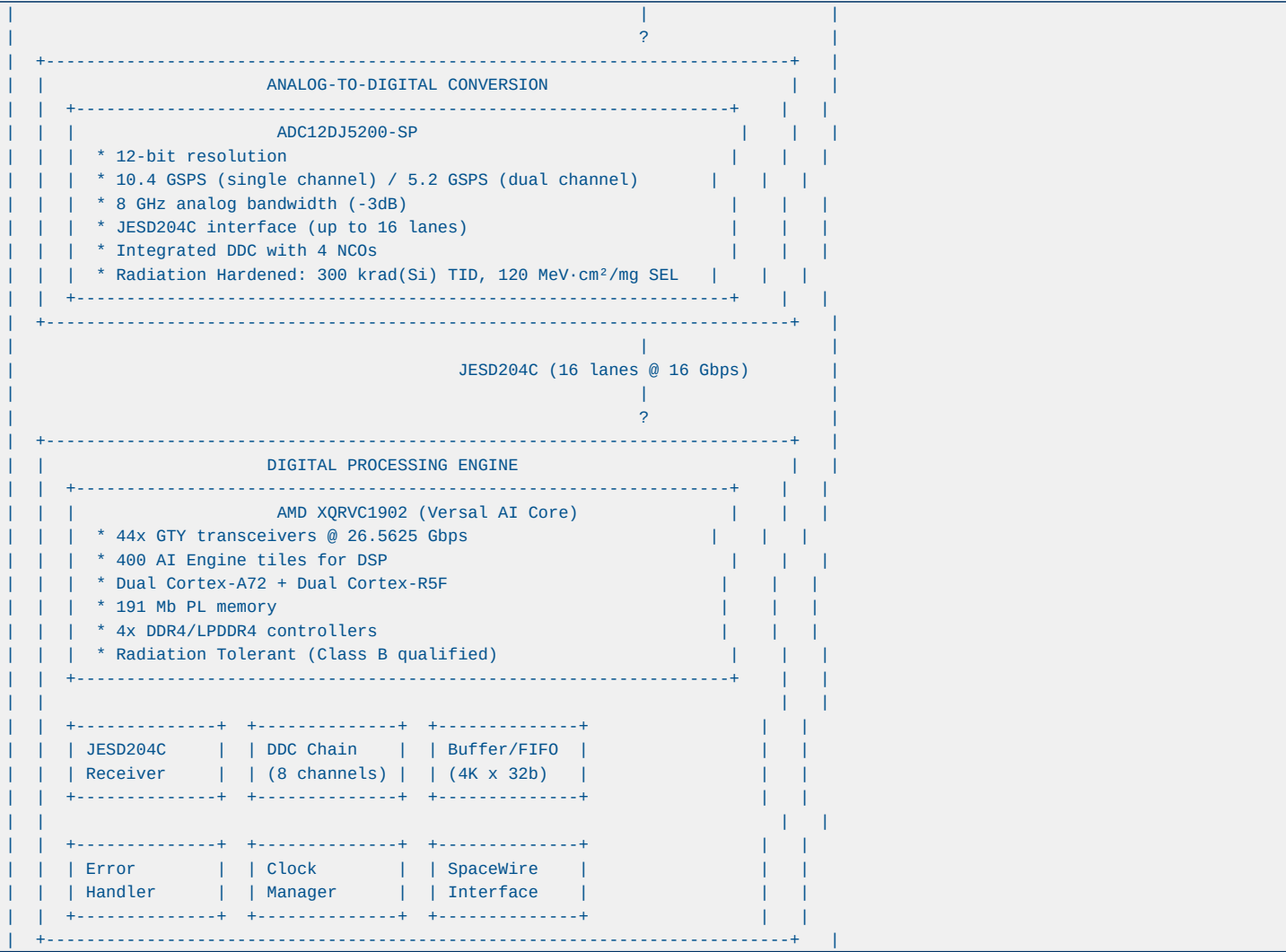
Option	Description	Pros	Cons	Selected
1.	Direct RF Sa	ADC samples X-b	Minimal compone	Requires 10+ GS **YES**
2.	Undersamplin	Bandpass sampli	Lower ADC speed	Complex aliasin No
3.	Downconversi	Analog mixer to	Better SNR, low	More components No
4.	Hybrid IF	Digital IF proc	Balanced approa	Complex archite No

3.2 Rationale for Direct RF Sampling

1. **Component Availability**: ADC12DJ5200-SP (TI) provides 10.4 GSPS with 8 GHz analog bandwidth

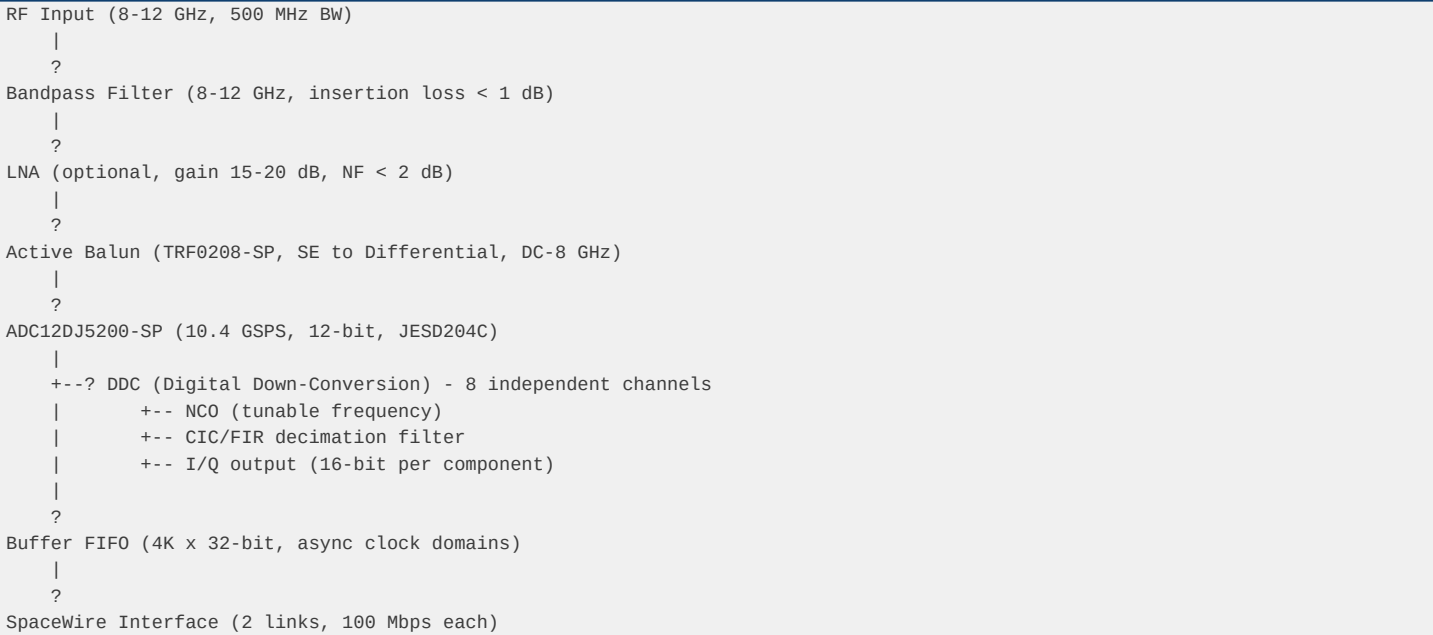
4. Functional Block Diagram





5. Data Flow Architecture

5.1 Signal Path



|
?
External Data Storage / Processing

5.2 Clock Distribution

VCX0 (122.88 MHz, <0.5 ps RMS jitter)
|
?
LMX2615-SP (PLL Synthesizer)
+--? 10.4 GHz Sampling Clock --? ADC CLK±
|
+--? Reference --? LMK04832-SP (Jitter Cleaner)
+--? Device Clock (DCLK) --? ADC, FPGA
+--? SYSREF --? ADC, FPGA (JESD204B/C sync)
+--? FPGA Clock --? FPGA Logic

5.3 Power Distribution

Input: 4.5V nominal (range 4.0-5.5V)
|
+--? TPS7H5002-SP (Buck) --? 1.0V @ 44A (FPGA Core)
|
+--? ISL70003ASEH (Buck) --? 1.8V @ 6A (FPGA Aux)
|
+--? ISL70003ASEH (Buck) --? 3.3V @ 3A (I/O, Clock)
|
+--? TPS7H1111-SP (LDO) --? 1.0V @ 1.5A (ADC AVDD)
|
+--? TPS7H1111-SP (LDO) --? 1.8V @ 1.5A (ADC DVDD)
|
+--? TPS7H1121-SP (LDO) --? 2.5V @ 2A (Clock, Balun)

Sequencing Order: 1.0V core -> 1.8V aux -> 3.3V I/O -> 1.0V analog -> 1.8V analog

6. Component Selection Summary

6.1 Critical Components

Component	Part Number	Function	Qualification	Rationale
ADC	ADC12DJ5200-SP	10.4 GSPS, 12-b	QML-V, 300 krad	Only space-grad
FPGA	XQRVC1902	Versal AI Core	Class B	44 GTY @ 26.5 G
Clock Synth	LMX2615-SP	10.4 GHz PLL	QMLV, 100 krad	45 fs RMS jitte
Clock Clean	LMK04832-SP	Jitter cleaner	QMLV, 100 krad	JESD204B/C SYSR
Balun	TRF0208-SP	SE to Diff	Space-grade	DC-8 GHz, match
Buck Ctrl	TPS7H5002-SP	Power regulator	QMLV-RHA, 100 k	Rad-hard, GaN-o
Buck Integ	ISL70003ASEH	9A Buck	QML Class V, 10	High efficiency
LDO	TPS7H1111-SP	Ultra-low noise	QMLV-RHA, 100 k	1.71 μVRMS, cri
Sequencer	TPS7H3014-SP	4-ch sequencer	QMLV-RHA, 100 k	Programmable, f

6.2 Alternative Components

Primary	Alternative	Qualification	Trade-off
ADC12DJ5200-SP	ADC12DJ5200-SEP	Rad-tolerant, 3	Lower TID, lowe
ADC12DJ5200-SP	EV12AQ600	Space-grade, 6.	Lower sample ra
XQRVC1902	XQRKU060	Class B, 12.5 G	GTH limited to
LMX2615-SP	LMX2694-SEP	Rad-tolerant, 3	Lower TID, smal
TPS7H5002-SP	RIC70847	Class V, Infine	Newer, limited

7. Design Constraints

7.1 Mechanical

- ****Dimensions****: 200 mm x 200 mm x 25 mm (max)

7.2 Electrical

- ****Input Voltage****: 4.5V nominal (4.0-5.5V range)

7.3 Environmental

- ****Operating Temp****: -40 degC to +85 degC

7.4 Radiation

- ****TID****: 100 krad(Si) minimum (2x margin for 50 krad mission)

8. Performance Budget

8.1 Signal Chain Performance

Parameter	Value	Notes
Input Frequency	8-12 GHz	X-band
Signal Bandwidth	500 MHz	Instantaneous
ADC Resolution	12-bit	Nominal
ENOB (at 10 GHz)	~8.8 bit	From datasheet
SNR (at 10 GHz)	~55.6 dB	From datasheet
SFDR (at 10 GHz)	~65 dBc	From datasheet
Clock Jitter	< 100 fs RMS	With LMK04832-S
DDC Decimation	2x to 32x	Software config
Output Data Rate	Up to 1.6 Gbps	After decimation

8.2 Power Budget

Block	Typical	Max	Notes
ADC12DJ5200-SP	4.0 W	4.5 W	From datasheet
FPGA XQRVC1902	60 W	80 W	Design dependent
Clock Subsystem	3.2 W	4.0 W	LMX2615 + LMK04
Power Management	5.0 W	7.0 W	Buck + LDO loss
Balun + Support	0.5 W	1.0 W	TRF0208-SP + misc
Total	**72.7 W**	**96.5 W**	With 30% margin

8.3 Thermal Budget

Parameter	Value	Notes
Max Junction Temp	125 degC	Per component spec
Ambient Temp	85 degC	Worst case
Thermal Resistance	1.5 degC/W	FPGA typical
Thermal Resistance	0.5 degC/W	With thermal interface
Thermal Resistance	2.0 degC/W	With heat sink
Max Power Dissipation	100 W	With active cooling

9. Reliability

9.1 MTBF Estimation

- ****Target MTBF****: > 100,000 hours (11.4 years)

9.2 Failure Modes

Failure Mode	Probability	Impact	Mitigation
ADC failure	Low	Loss of acquisition	Watchdog, reset
FPGA SEU	Medium	Bit flip, logic error	TMR, EDAC, scrub

Clock unlock	Low	Loss of samplin	Redundant clock
Power supply fa	Low	Module shutdown	Current limitin
SEL event	Very Low	Latch-up, damag	Current limitin

10. Interface Control Document (ICD) Summary

10.1 External Interfaces

Interface	Type	Pins	Description
RF Input	SMA	2 (signal + gnd)	8-12 GHz input
Power Input	MIL-DTL-38999	4 (VIN, GND, GN)	4.5V, 25A max
SpaceWire	MIL-DTL-38999	8 (2 links)	Data + Strobe,
JTAG	Box header	14	FPGA configurat
Debug	Box header	10	UART, SPI, I2C
Test Points	Via	10	Power rails, cl

10.2 Internal Interfaces

Interface	Standard	Width	Clock	Description
ADC to FPGA	JESD204C	16 lanes	10.4 GHz	64B/66B encodin
FPGA to DDR4	DDR4	256-bit	1200 MHz	Memory interfac
FPGA to SpaceWi	SWS	2 links	100 Mbps	Data output
FPGA to Clock	SPI	24-bit	10 MHz	Configuration
FPGA to Power	I2C	8-bit	400 kHz	Monitoring

11. Verification Approach

11.1 Verification Methods

- **Analysis**: Worst-case analysis (WCA), Parts stress analysis (PSA)

11.2 Test Points

Test Point	Location	Signal	Purpose
TP1	Power input	VIN	Input voltage m
TP2	1.0V core	VCCINT	FPGA core volta
TP3	1.8V aux	VCCAUX	FPGA aux voltage
TP4	3.3V I/O	VCCO	I/O voltage
TP5	1.0V analog	AVDD	ADC analog supp
TP6	1.8V digital	DVDD	ADC digital sup
TP7	Clock output	CLK	Sampling clock
TP8	SYSREF	SYSREF	JESD204 sync
TP9	JESD lane 0	TX0	Data lane monit
TP10	Error flag	ERR	System error

12. Risk Assessment Summary

Risk ID	Description	Probability	Impact	Mitigation	Status
R001	Clock jitter ex	Low	High	Use LMX2615-SP	Open
R002	JESD204C lane m	Low	Medium	GTY @ 26.5 Gbps	Open
R003	Power sequencin	Medium	High	TPS7H3014-SP wi	Open
R004	TID accumulatio	Low	Medium	100+ krad compo	Open
R005	SEL event	Very Low	High	All components	Open
R006	Thermal hotspot	Medium	Medium	Thermal vias, c	Open
R007	Component obsol	Low	Medium	Multi-source, a	Open

13. Documents Cross-Reference

Document	ID	Description
System Architec	XBDM-DOC-001	This document
Power Tree	XBDM-DOC-002	Detailed power
Clock Tree	XBDM-DOC-003	Clock distribut
Risk Assessment	XBDM-DOC-004	Detailed risk m
Component Selec	XBDM-DOC-005	Component ratio
WCA	XBDM-DOC-006	Worst-case anal
PSA	XBDM-DOC-007	Parts stress an
ICD Mechanical	XBDM-DOC-008	Mechanical inte
ICD VHDL	XBDM-DOC-009	Digital interfa
Test Plan	XBDM-DOC-010	Verification an

14. Revision History

Version	Date	Author	Description
1.0	2026-09-09	System Engineer	Initial release
